

Question

The amount of water required to make 10% ethanol in water solution using 10 mL of pure ethanol is:

- A** 90 mL
- B** 50 mL
- C** 30 mL
- D** 60 mL

Question

A solution is obtained by mixing 300 g of 25% solution and 400 g of 40% solution by mass. Calculate the mass percentage of solute in the resulting solution?

A 40.35%

B 33.57%

C 10.23%

D 18.28%

Question

What is the percentage of solute in the resultant solution, if is obtained by mixing 300 g of 30% and 200 g of 20% solution by weight?

A 50%

B 26%

C 62%

D 32%

Question

What will be the molarity of pure water?

- A** 18 M
- B** 50.0 M
- C** 55.6 M
- D** 100 M

Question

A sample of nitric acid is 69% by mass and it has a concentration of 15.44 moles per litre. Its density is:

- A** 1.86 g/cc
- B** 1.41 g/cc
- C** 2.60 g/cc
- D** 1.02 g/cc

Question

What volume of water is to be added to 100 cm^3 of 0.5 M NaOH solution to make it 0.1 M solution?

- A** 200 cm^3
- B** 400 cm^3
- C** 500 cm^3
- D** 100 cm^3

Question

Match the following physical quantities with their units and choose the correct codes from the options given below.

A **a** **b** **c** **d**
 3 2 1 4

B 2 1 4 3

C 4 2 1 3

D 1 2 4 3

Column I (Physical quantity)		Column II (Unit)	
a.	Molarity	1.	Mol
b.	Mole fraction	2.	Unitless
c.	Mole	3.	mol L^{-1}
d.	Molality	4.	mol kg^{-1}

Question

250 g solution of D-glucose in water contains 10.8% of carbon by weight.

The molality of the solution is nearest to

(Given: Atomic weights are, H = 1u; C = 12u; O = 16u)

A 1.03

B 2.06

C 3.09

D 5.40

Question

4 g equimolar mixture of NaOH and Na_2CO_3 contains x g of NaOH and y g of Na_2CO_3 . The value of x is _____ g. (Nearest Integer)

Question



Consider the above reaction, what mass of CaCl_2 will be formed if 250 mL of 0.76 M HCl reacts with 1000 g of CaCO_3 ?

(Given: Molar mass of Ca, C, O, H and Cl are 40, 12, 16, 1 and 35.5 g mol^{-1} , respectively)

- A** 3.908 g
- B** 2.636 g
- C** 10.545 g
- D** 5.272 g

Question

In the balanced chemical reaction,
 $\text{IO}_3^- + a \text{I}^- + b \text{H}^+ \rightarrow c \text{H}_2\text{O} + d \text{I}_2$
a, b, c and d respectively correspond to

A 5, 6, 3, 3

B 5, 3, 6, 3

C 3, 5, 3, 6

D 5, 6, 5, 5

Question

Only 2 mL of KMnO_4 solution of unknown molarity is required to reach the end point of a titration of 20 mL of oxalic acid (2 M) in acidic medium. The molarity of KMnO_4 solution should be _____ M.

Question

When 10 mL of an aqueous solution of KMnO_4 was titrated in acidic medium, an equal volume of 0.1 M of an aqueous solution of ferrous sulphate was required for complete discharge of colour. The strength of KMnO_4 in grams per litre is _____ $\times 10^{-2}$. (Nearest integer)
[Atomic mass of K = 39, Mn = 55, O = 16]

Question

10.0 ml of 0.05 M KMnO_4 solution was consumed in a titration with 10.0 mL of given oxalic acid dehydrate solution. The strength of given oxalic acid solution is

_____ $\times 10^{-2}$ g/L

(Round off to the Nearest Integer).

Question

10.0 mL of Na_2CO_3 solution is titrated against 0.2 M HCl solution. The following titre values were obtained in 5 readings:

4.8 mL, 4.9 mL 5.0 mL 5.0 mL and 5.0 mL

Based on these readings, and convention of titrimetric estimation, the concentration of Na_2CO_3 solution is _____ mM. (Round off to the nearest integer).

Question

When 10 ml of an aqueous solution of Fe^{2+} ions was titrated in the presence of dil H_2SO_4 using diphenylamine indicator, 15 mL of 0.02 M solution of $\text{K}_2\text{Cr}_2\text{O}_7$ was required to get the end point. The molarity of the solution containing Fe^{2+} ions is $x \times 10^{-2}$ M. The value of x is. (Nearest integer)

Question

Homogeneous mixture of two or more than two components is called

- A** Solute
- B** Solvent
- C** Both (A) and (B)
- D** Solution

Question

Which of the following possesses physical state of solute and solvent as liquid and solid respectively?

- A** solutions of sugar in water
- B** zinc amalgam
- C** solutions of Naphthalene in benzene
- D** brass

Question

In amalgam of mercury with sodium, solvent is

- A** Mercury
- B** Sodium
- C** Amalgam
- D** None of these

Question

Match List-I with List-II to find out the correct option.

- A** A-V, B-III, C-IV, D-II, E-I
- B** A-II, B-I, C-III, D-VI, E-IV
- C** A-VI, B-IV, C-III, D-II, E-I
- D** A-I, B-VI, C-IV, D-II, E-III

List-I		List-II	
(A)	Soda water	(I)	A solution of gas in solid
(B)	Sugar solution	(II)	A solution of gas in gas
(C)	German silver	(III)	A solution of solid in liquid
(D)	Air	(IV)	A solution of solid in solid
(E)	Hydrogen gas in palladium	(V)	A solution of gas in liquid
		(VI)	A solution of liquid in solid

Question

Which of the following pair of solution having different physical state of solute?

- A** homogenous mixture of chloroform in N_2 gas, solution of CO_2 in water
- B** brass, homogenous mixture of camphor in N_2 gas
- C** sodium amalgam, moist air
- D** solution of H_2 gas in Pd metal, mixture of N_2 and O_2

Question

Which of the following alternative is correct for physical state of solute and solvent for a solution of camphor in N_2 gas?

- A** solid, gas
- B** solid, liquid
- C** gas, solid
- D** gas, gas

Question

According to Henry's law, the solubility of a gas in a given volume of liquid increases with increase in:

- A** Temperature
- B** Pressure
- C** Both (A) and (B)
- D** None of these

Question

Some of the following gases are soluble in water due to formation of their ions:

I. CO_2

II. NH_3

III. HCl

IV. CH_4

V. H_2

Water insoluble gases can be:

A I, IV, V

B I, V

C I, II, III

D IV, V

Question

The partial pressure of ethane over a solution containing 6.56×10^{-3} g of ethane is 1 bar. If the solution contains 5.00×10^{-2} g of ethane, then what shall be the partial pressure of the gas?

Question

The solubility of $\text{N}_2(\text{g})$ in water exposed to the atmosphere, when its partial pressure is 593 mm is 5.3×10^{-4} M. Its solubility at 760 mm and at the same temperature is:

- A** 4.1×10^{-4} M
- B** 6.8×10^{-4} M
- C** 1500 M
- D** 2400 M

Question

Which one of the following statements regarding Henry's law is not correct?

- A** Different gases have different K_H (Henry's law constant) values at the same temperature.
- B** The value of K_H increases with increase of temperature and K_H is a function of the nature of the gas.
- C** The partial pressure of the gas in vapour phase is proportional to the mole fraction of the gas in the solution.
- D** Higher the value of K_H at a given pressure, higher is the solubility of the gas in the liquid.

Question

The solubility of gas in liquid depends upon

- A** The nature of gas
- B** The temperature
- C** The nature of solvent
- D** All of the above

Question

Low concentration of oxygen in the blood and tissues of people living at high altitude is due to.....

- A** low temperature
- B** low atmospheric pressure
- C** high atmospheric pressure
- D** both low temperature and high atmospheric pressure

Question

The value of Henry's constant, K_H is.....

- A** greater for gases with higher solubility
- B** greater for gases with lower solubility
- C** constant for all gases
- D** not related to the solubility of gases

Question

Fishes feel uncomfortable in warm water due to

- A** fishes do not like warmness
- B** higher amount of impurities
- C** low solubility of oxygen at higher temperature
- D** greater population of fishes

Question

The gas for which Henry's law is not applicable in aqueous solution is:

A HCl

B N₂

C Xe

D He

Question

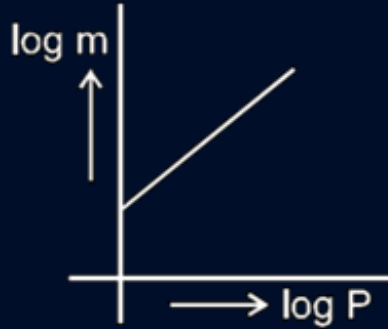
Henry's law is not applicable at

- A** low pressure
- B** high pressure
- C** gas does not react with liquid
- D** high temperature

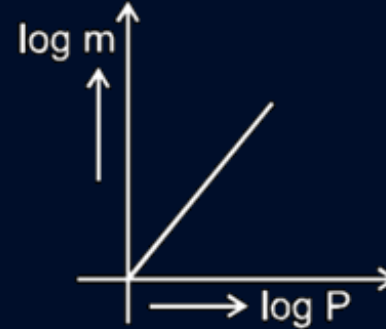
Question

Which of the following curves represents the Henry's law?

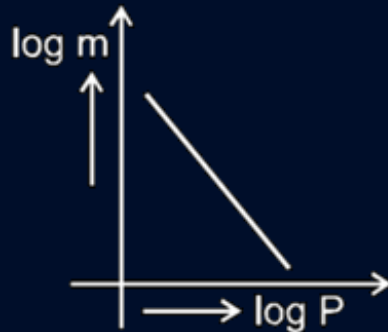
A



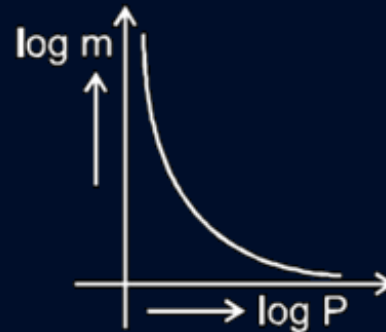
B



C



D



Question

If N_2 gas is bubbled through water at 293 K, how many millimoles of N_2 gas would dissolve in 1 litre of water? Assume that N_2 exerts a partial pressure of 0.987 bar. Given that Henry's law constant for N_2 at 293 K is 76.48 kbar.

Question

If O_2 gas is bubbled through water at 303 K, the number of millimoles of O_2 gas that dissolve in 1 litre of water is _____.

(Given: Henry's Law constant for O_2 at 303 K is 46.82 k bar and partial pressure of $O_2 = 0.920$ bar)

(Assume solubility of O_2 in water is too small, nearly negligible)

Question

The oxygen dissolved in water exerts a partial pressure of 20 kPa in the vapour above water. The molar solubility of oxygen in water is ____ $\times 10^{-5}$ mol dm⁻³.

[Given: Henry's law constant = $K_H = 8.0 \times 10^4$ kPa for O₂ Density of water with dissolved oxygen = 1.0 kg dm⁻³]

Question

A company dissolves 'X' amount of CO_2 at 298 K in 1 litre of water to prepare soda water. $X = ___ \times 10^{-3}\text{g}$. (Given: partial pressure of CO_2 at 298 K = 0.835 bar. Henry's law constant for CO_2 at 298 K = 1.67 kbar. Atomic mass of H, C and O is 1, 12 and 16 g mol^{-1} , respectively)

Question

Henry's Law Constant for CO_2 in water is $1.67 \times 10^8 \text{ P}_a$ at 298 K. Calculate the quantity of CO_2 in 1 L of soda water when packed under 2.5 atm CO_2 pressure at 298 K.

Question

The Henry's law constant for the solubility of nitrogen gas in water at 298 K is 1×10^5 atm. The mole fraction of nitrogen in air is 0.8. Calculate the number of moles of nitrogen from air dissolved in 10 moles of water at 298 K and 5 atm pressure of air.

Question

Which of the following statements is correct, if the intermolecular forces in liquids A, B and C are in the order $A < B < C$?

- A** B evaporates more readily than A
- B** B evaporates less readily than C
- C** A and B evaporate at the same rate
- D** A evaporates more readily than C

Question

Vapour pressure is the pressure exerted by vapours

- A** In equilibrium with liquid
- B** In any condition
- C** In an open system
- D** In atmospheric conditions

Question

The boiling point of C_6H_6 , CH_3OH , $\text{C}_6\text{H}_5\text{NH}_2$ and $\text{C}_6\text{H}_5\text{NO}_2$ are 80°C , 65°C , 184°C and 212°C respectively. Which will show highest vapour pressure at room temperature-

- A** C_6H_6
- B** CH_3OH
- C** $\text{C}_6\text{H}_5\text{NH}_2$
- D** $\text{C}_6\text{H}_5\text{NO}_2$

Question

Boiling point of water is defined as the temperature at which–

- A** Vapour pressure of water becomes equal to that of atmospheric pressure
- B** Bubbles are formed
- C** Steam comes out
- D** None of the above

Question

Unit of Henry's constant K_H is same as that of

- A** Volume
- B** Temperature
- C** Energy
- D** Pressure

Question

Gases with their Henry's constant values are given. The gas having maximum solubility will be

Gas A $K_H = 21.2 \text{ k bar}$

Gas C $K_H = 5.6 \text{ k bar}$

Gas B $K_H = 11.2 \text{ k bar}$

Gas D $K_H = 2.4 \text{ k bar}$

A A

B B

C C

D D

Question

If partial pressure of Gas A = 0.4 bar, Gas B = 0.2 bar and Gas C = 0.5 bar, then the gas having maximum solubility (neglecting other factors) is

- ☐ A A
- ☐ B B
- ☐ C C
- ☐ D All are equal soluble

Question

Henry's law constant for CO_2 in water is $1.67 \times 10^8 \text{ Pa}$ at 298 K. Calculate the quantity of CO_2 in 500 mL of soda water when packed under 2.5 atm CO_2 pressure at 298 K.

Question

The solubility of N_2 in water is 2.2×10^{-4} g in 100 g of H_2O at 20°C when the pressure of N_2 over the solution is 1.2 atm. Calculate the solubility at 20°C when the pressure of N_2 over the solution is 10 atm.

Question

The partial pressure of O_2 in air is 0.21 atm at sea level. If aquatic life requires a concentration of O_2 at least 4.16 mg/litre, will the aquatic life sustain in sea level lake. K_H for O_2 is $1.3 \times 10^{-3} \text{ mol litre}^{-1} \text{ atm}^{-1}$.

Question

H_2S , a toxic gas with rotten egg like smell, is used for the qualitative analysis. If the solubility of H_2S in water at STP is 0.195 m, calculate Henry's law constant.

Question

The solubility of N_2 at 20°C is $5.6 \times 10^{-4} \text{ mol litre}^{-1}$ from air (80% N_2 + 20% O_2) at 1 atm. A deep sea diver inside (at 20°C) sea breaths air at 5 atm. If total volume of blood is 6.0 litre. After some time the diver comes on the water surface where air pressure is 1 atm. Calculate how much volume of N_2 will come out from the blood at these 1 atm P and 20°C conditions.

Question

At the sea level, the dry air mass percentage composition is given as nitrogen gas : 70.0, oxygen gas : 27.0 and argon gas : 3.0. If total pressure is 1.15 atm, then calculate the ratio of following respectively:

- (i) partial pressure of nitrogen gas to partial pressure of oxygen gas
- (ii) partial pressure of oxygen gas to partial pressure of argon gas (Given : Molar mass of N, O and Ar are 14, 16, and 40 g mol⁻¹ respectively)

A 4.26, 19.3

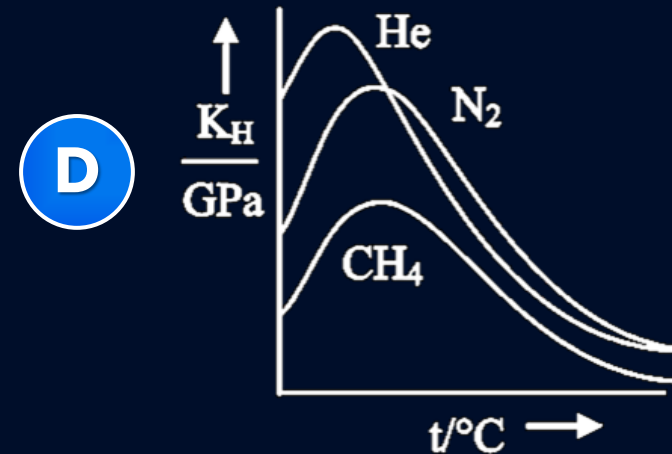
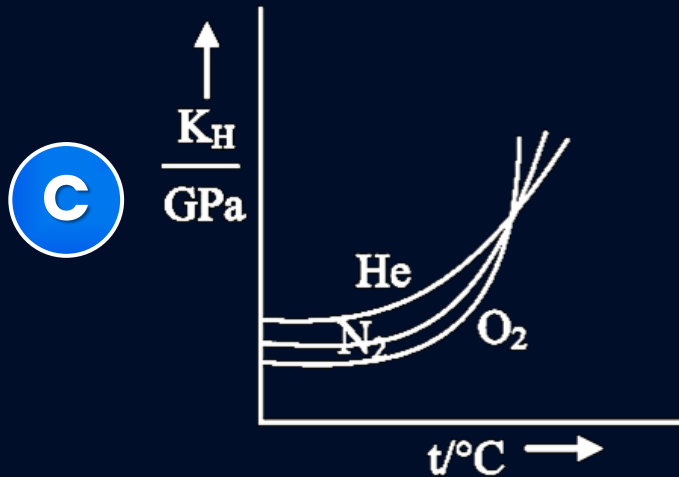
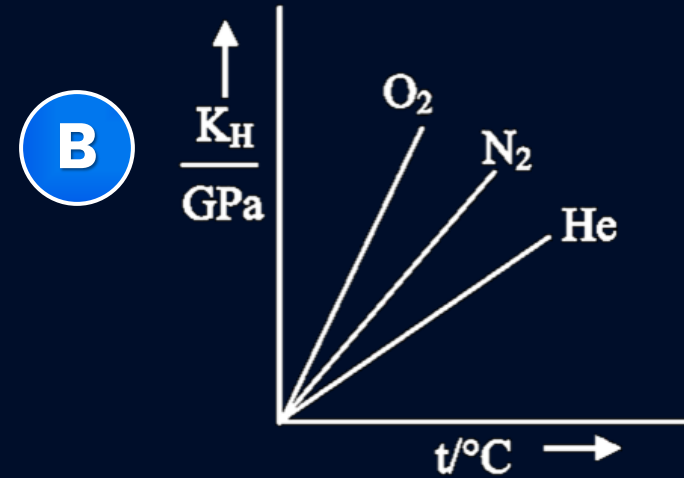
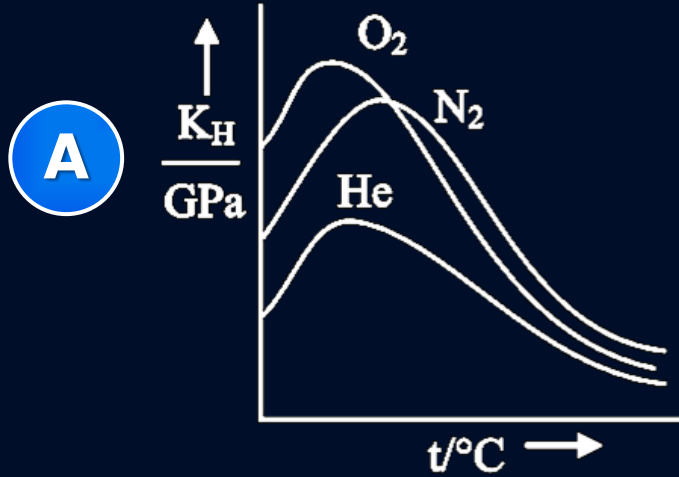
B 2.59, 11.85

C 5.46, 17.8

D 2.96, 11.2

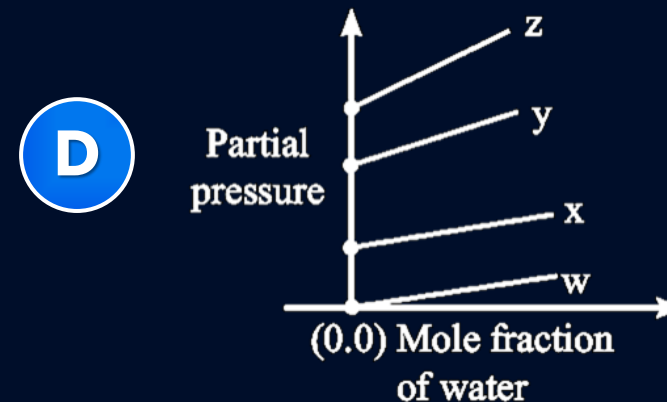
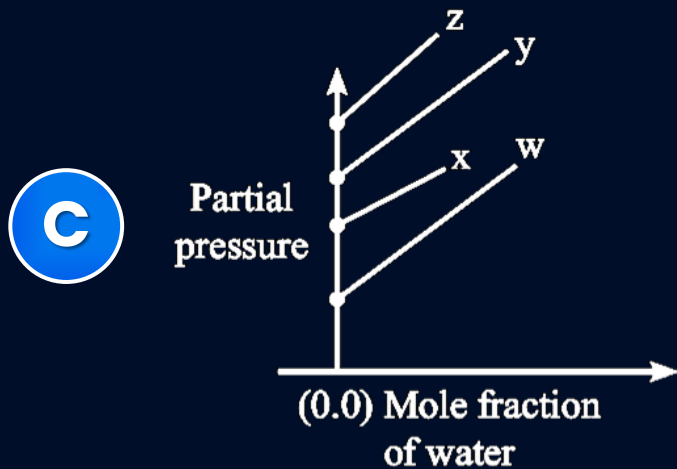
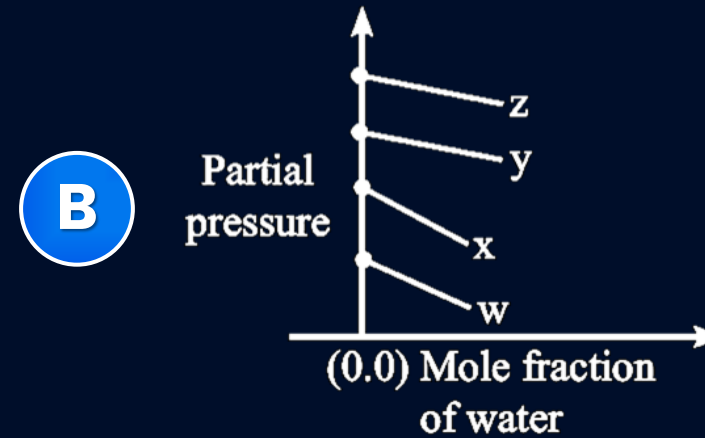
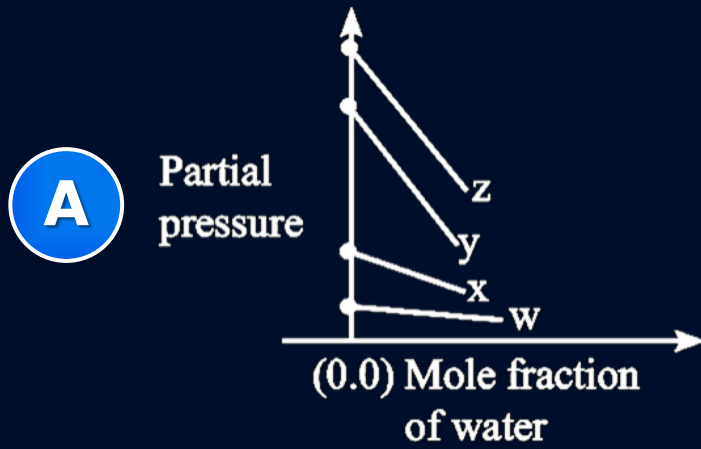
Question

Which of the following graph correctly represents the plots of K_H at 1 bar gases in water versus temperature?



Question

For the solution of the gases w, x, y and z in water at 298 K, the Henry's law constants (K_H) are 0.5, 2, 35 and 40 k bar respectively. The correct plot for the given data is



Question

Liquid A and B form an ideal solution. If vapour pressure of pure A and B are 50 Nm^{-2} & 200 Nm^{-2} respectively, the vapour pressure of a solution of A in B containing 0.2 mole fraction of A would be:

- A** 700 Nm^{-2}
- B** 300 Nm^{-2}
- C** 260 Nm^{-2}
- D** 140 Nm^{-2}

Question

The vapour pressure of two pure liquids (A) and (B) are 100 and 80 torr respectively. The total pressure of solution obtained by mixing 2 mole of (A) and 3 mole of (B) would be

A 120 torr

B 36 torr

C 88 torr

D 180 torr

Question

Two liquids A and B form an ideal solution. At 300 K the vapour pressure of solution of 1 mole of A and x moles of B is 550 mm. If the vapour pressures of pure A and B are 400 mm and 600 mm respectively, then x is-

- A** 1
- B** 2
- C** 3
- D** 4

Question

The Total pressure observed by mixing two liquid A and B is 350 mm Hg when their mole fractions are 0.7 and 0.3 respectively.

The Total pressure becomes 410 mm Hg if the mole fractions are changed to 0.8 and 0.2 respectively for A and B. The vapour pressure of pure A is mmHg.

Question

At 300 K, the vapour pressure of an ideal solution containing one mole of A and 3 mole of B is 550 mm of Hg. At the same temperature, if one mole of B is added to this solution, the vapour pressure of solution increases by 10 mm of Hg. Calculate the vapour pressure of A and B (in mm Hg) respectively in their pure state.

- A** 400, 600
- B** 100, 200
- C** 300, 400
- D** 800, 400

Question

At a given temperature, total vapour pressure in Torr of a mixture of volatile components A and B is given by

$$P = 120 - 75 X_B$$

Hence, vapour pressure of pure A and B respectively (in Torr) are-

- A** 120, 75
- B** 120, 195
- C** 120, 45
- D** 75, 45

Question

At 40°C, the vapor pressure (in torr) of methyl alcohol (A) and ethyl alcohol (B) solution is represented by:

$$P = 120 X_A + 138;$$

where X_A is mole fraction of methyl alcohol. The value of vapor pressure of methyl and ethyl alcohol respectively and are:

- A** 138, 258
- B** 258, 138
- C** 120, 138
- D** 138, 125

Question

The vapour pressures of pure liquids A and B are 400 and 600 mm Hg respectively at 298 K. On mixing the two liquids, the sum of their initial volumes is equal to the volume of the final mixture. The mole fraction of liquid B is 0.5 in the mixture. The vapour pressure of the final solution, the mole fractions of components A and B in vapour phase respectively, are

- A** 500 mm Hg, 0.4, 0.6
- B** 500 mm Hg, 0.5, 0.5
- C** 450 mm Hg, 0.4, 0.6
- D** 450 mm Hg, 0.5, 0.5

Question

A mixture of ethyl alcohol and propyl alcohol has a vapour pressure of 290 mm at 300 K. The vapour pressure of propyl alcohol is 200 mm. If the mole fraction of ethyl alcohol is 0.6, its vapour pressure (in mm) at the same temperature will be

- A** 350
- B** 300
- C** 700
- D** 360

Question

The correct option for the value of vapour pressure of a solution at 45°C with benzene to octane in molar ratio 3 : 2 is

[At 45°C vapour pressure of benzene is 280 mm Hg and that of octane is 420 mm Hg]

- A** 336 mm of Hg
- B** 350 mm of Hg
- C** 160 mm of Hg
- D** 168 mm of Hg

Question

Consider a binary solution of two volatile liquid components 1 and 2. x_1 and y_1 are the mole fractions of component 1 in liquid and vapour phase, respectively.

The slope and intercept of the linear plot of $\frac{1}{x_1}$ vs $\frac{1}{y_1}$ are given respectively as:

A $\frac{P_1^0}{P_2^0}, \frac{P_2^0 - P_1^0}{P_2^0}$

B $\frac{P_1^0}{P_2^0}, \frac{P_1^0 - P_2^0}{P_2^0}$

C $\frac{P_2^0}{P_1^0}, \frac{P_1^0 - P_2^0}{P_2^0}$

D $\frac{P_2^0}{P_1^0}, \frac{P_2^0 - P_1^0}{P_2^0}$

Question

The vapor pressure of pure benzene and toluene are 160 and 60 torr respectively. The mole fraction of toluene in vapor phase in contact with an equimolar solution of benzene and toluene is:

- A** 0.50
- B** 0.6
- C** 0.27
- D** 0.73

Question

An open beaker of water in equilibrium with its water vapour is in a sealed container. When a few grams of glucose are added to the beaker of water, the rate at which water molecules

- A** Leaves the solution increases
- B** Leaves the vapour increases
- C** Leaves the vapour decreases
- D** Leaves the solution decreases

Question

A solution of two miscible liquids showing negative deviation from Raoult's law will have:

- A** increased vapour pressure, increased boiling point
- B** increased vapour pressure, decreased boiling point
- C** decreased vapour pressure, decreased boiling point
- D** decreased vapour pressure, increased boiling point

Question

Liquid A and B form an ideal solution. The vapour pressures of pure liquids A and B are 350 and 750 mm Hg respectively at the same temperature. If x_A and x_B are the mole fraction of A and B in solution while y_A and y_B are the mole fraction of A and B in vapour phase then:

A $\frac{x_A}{x_B} < \frac{y_A}{y_B}$

B $\frac{x_A}{x_B} = \frac{y_A}{y_B}$

C $\frac{x_A}{x_B} > \frac{y_A}{y_B}$

D $(x_A - y_B) < (x_B - y_B)$

Question

Identify the mixture that shows positive deviations from Raoult's Law

- A** $(\text{CH}_3)_2\text{CO} + \text{C}_6\text{H}_5\text{NH}_2$
- B** $\text{CHCl}_3 + \text{C}_6\text{H}_6$
- C** $\text{CHCl}_3 + (\text{CH}_3)_2\text{CO}$
- D** $(\text{CH}_3)_2\text{CO} + \text{CS}_2$

Question

On mixing 10 mL of acetone with 50 mL of chloroform, the total volume of the solution is-

- A** < 60 mL
- B** > 60 mL
- C** $= 60$ mL
- D** unpredictable

Question

Total vapor pressure of mixture of 1 mol A ($p_A^\circ = 150$ torr) and 2 mol B ($p_B^\circ = 240$ torr) is 200 torr. In this case-

- A** there is positive deviation from Raoult's law
- B** there is negative deviation from Raoult's law
- C** there is no deviation from Raoult's law
- D** molecular masses of A and B are also required for calculating the deviation

Question

Which of the following liquid pairs shows a positive deviation from Raoult's law?

- A** Water-hydrochloric acid
- B** Water-nitric acid
- C** Acetone-chloroform
- D** Benzene-ethanol

Question

100 ml of liquid A and 25 ml of liquid B is mixed to give a solution which does not obey Raoult's law. The volume of the solution-

- A** will be 125 ml
- B** can be $>$ or $<$ than 125 ml
- C** can be greater than, equal to or less than 125 ml
- D** will be less than 125 ml

Question

An azeotropic solution of two liquids has boiling point lower than either when it-

- A** shows a negative deviation from Raoult's law
- B** shows a positive deviation from Raoult's law
- C** shows no deviation from Raoult's law
- D** is saturated

Question

H₂O and perchloric acid (b.p. 383 K) form constant boiling mixture at 71.6% of perchloric acid. The boiling point of the solution at this composition is-

- A** > 373 but < 383 K
- B** < 373 K
- C** $>$ than 383 K
- D** $= 373$ K

Question

Azeotropic mixture of water and HCl boils at 381.5 K. By distilling the mixture it is possible to obtain-

- A** Pure HCl only
- B** Pure water only
- C** Neither HCl nor water
- D** Both water and HCl in pure state

Question

Which of the following binary mixture does not show the behaviour of minimum boiling azeotropes?

- A** $\text{H}_2\text{O} + \text{CH}_3\text{COC}_2\text{H}_5$
- B** $\text{C}_6\text{H}_5\text{OH} + \text{C}_6\text{H}_5\text{NH}_2$
- C** $\text{CS}_2 + \text{CH}_3\text{COCH}_3$
- D** $\text{CH}_3\text{OH} + \text{CHCl}_3$

Question

Match List-I with List-II

Choose the correct answer from the options given below:

A (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

B (A)-(II), (B)-(IV), (C)-(I), (D)-(III)

C (A)-(III), (B)-(IV), (C)-(I), (D)-(II)

D (A)-(II), (B)-(I), (C)-(IV), (D)-(III)

List-I		List-II	
(A)	Solution of chloroform and acetone	(I)	Minimum boil azeotrope
(B)	Solution of ethanol and water	(II)	Dimerizes
(C)	Solution of benzene and toluene	(III)	Maximum boi azeotrope
(D)	Solution of acetic acid in benzene	(IV)	$\Delta V_{\text{mix}} = 0$

Question

The mixture that forms maximum boiling azeotrope is

- A** Ethanol + Water
- B** Acetone + Carbon disulphide
- C** Heptane + Octane
- D** Water + Nitric acid

Question

A binary liquid solution is prepared by mixing n-heptane and ethanol. Which one of the following statements is correct regarding the behaviour of the solutions?

- A** The solution is non-ideal, showing -ve deviation from Raoult's law
- B** n-heptane shows +ve deviation while ethanol shows -ve deviation from Raoult's law
- C** The solutions formed is an ideal solution.
- D** The solutions is non-ideal, showing +ve deviation from Raoult's law

Question

The vapour pressure of solution containing 5 g of non-electrolyte in 90 g of water at a particular temperature is 2985 Nm^{-2} . The vapour pressure of pure water at that temperature is 3000 Nm^{-2} . The molecular weight of the solute is–

- A** 180
- B** 90
- C** 270
- D** 200

Question

For a solution containing non-volatile solute, the relative lowering of vapour pressure is 0.2. If the solution contains 5 moles in all, which of the following are true?

- I. Mole fraction of solute in the solution is 0.2
- II. No. of moles of solute in the solution is 0.2
- III. No. of moles of solvent in the solution is 4
- IV. Mole fraction of solvent is 0.2

- A** I, IV
- B** II, III
- C** I, III
- D** II, IV

Question

Lowering of vapour pressure of an aqueous solution of a non-volatile, non-electrolyte 1 M aqueous solution at 100°C is

- A** 14.12 torr
- B** 312 torr
- C** 13.45 torr
- D** 352 torr

Question

The vapour pressure lowering caused by the addition of 100 g of sucrose (molecular mass = 342 g mol^{-1}) to 1000 g of water, if the vapour pressure of water at 25°C is 23.8 mm of Hg is

- A** 0.12 mm Hg
- B** 0.125 mm Hg
- C** 1.15 mm Hg
- D** 1.25 mm Hg

Question

Mass of Urea (NH_2CONH_2) required to be dissolved in 1000 g of water to reduce the vapour pressure of water by 25% is _____ g. (Nearest Integer)

Given: Molar mass of N, C, O and H are 14, 12, 16 and 1 g mol⁻¹ respectively.

Question

The vapour pressure of 30 % (w/v) aqueous solution of glucose is _____ mm Hg at 25°C.

[Given: The density of 30%(w/v), aqueous solution of glucose is 1.2 g cm^{-3} and vapour pressure of pure water is 24 mmHg.]

(Molar mass of glucose is 180 g mol^{-1} .)

Question

At room temperature, a dilute solution of urea is prepared by dissolving 0.60 g of urea in 360 g of water. If the vapour pressure of pure water at this temperature is 35 mm Hg, lowering of vapour pressure will be: (molar mass of urea = 60 g mol^{-1})

- A** 0.027 mmHg
- B** 0.031 mmHg
- C** 0.028 mmHg
- D** 0.017 mmHg

Question

What weight of glucose must be dissolved in 100 g of water to lower the vapour pressure by 0.20 mm Hg? (Assume dilute solution is being formed)

Given: Vapour pressure of pure water is 54.2 mm Hg at room temperature. Molar mass of glucose is 180 g mol^{-1}

A 4.69 g

B 3.59 g

C 2.59 g

D 3.69 g

Question

The vapour pressure of a solvent decreased by 10 mm of Hg when a non-volatile solute was added to the solvent. The mole fraction of solute in solution is 0.2. What would be the mole fraction of solvent if decrease in vapour pressure is 20 mm of Hg?

- A** 0.8
- B** 0.6
- C** 0.4
- D** 0.2

Question

If at certain temperature, the vapour pressure of pure water is 25 mm of Hg and that of a very dilute aqueous urea solution is 24.5 mm of Hg, the molality of solution is

- A** 0.02
- B** 1.2
- C** 1.11
- D** 0.08

Question

What does point A and B represent in the following diagram?

Point A

A Boiling point of solvent

B Boiling point of solution

C Boiling point of solute

D Boiling point of solvent

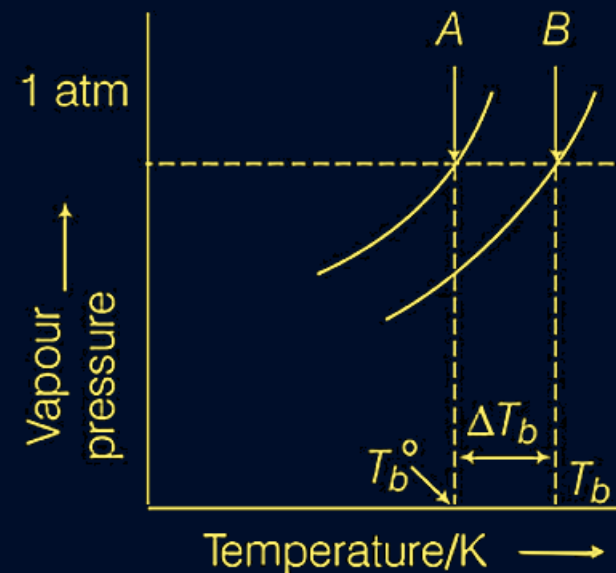
Point A

Boiling point of solution

Boiling point of solvent

Boiling point of solvent

Boiling point of solute



Question

30 g of urea is present in 100 g of water. Then find out boiling point of solution in $^{\circ}\text{C}$?

Question

If the elevation in boiling point of a solution of 10 gm of solute (mol. Wt. = 100) in 100 gm of water is T_b , the ebullioscopic constant of water is-

- A** 10
- B** $10 T_b$
- C** T_b
- D** $T_b/10$

Question

At certain Hill-station pure water boils at 99.725°C . If K_b for water is $0.513^{\circ}\text{C kg mol}^{-1}$. The boiling point of 0.69 m solution of urea will be-

- A** 100.079°C
- B** 103°C
- C** 100.359°C
- D** Unpredictable

Question

To observe an elevation of boiling point is 0.05°C , the amount of a solute (molecular weight = 100) to be added to 100 g of water ($K_b = 0.5$) is

- A** 2 g
- B** 0.05 g
- C** 1 g
- D** 0.75 g

Question

If molality of the dilute solution is doubled, the value of molal depression constant $[K_f]$ will be

- A** doubled
- B** halved
- C** tripled
- D** unchanged

Question

How much gram of glycol is dissolved in 5L H_2O so that freezing point of solution becomes -6°C . (K_f of water = $1.86 \text{ Kg K mol}^{-1}$).

Question

The freezing point of an aqueous solution is -1.5°C . Then find the boiling point of the solution.

Question

The molar freezing point constant for water is $1.86^{\circ}\text{C}/\text{mole}$. If 342 gm of cane sugar ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$) is dissolved in 1000 gram of water the solution will freeze at-

- A** -1.86°C
- B** 1.86°C
- C** -3.92°C
- D** 2.42°C

Question

Y g of non-volatile organic substance of molecture mass M is dissolved in 250 benzene. Molal elevation constant of benzene is K_b . Elevation in its boiling point is given by

- A** $\frac{M}{K_b Y}$
- B** $\frac{4K_b Y}{M}$
- C** $\frac{K_b Y}{4M}$
- D** $\frac{K_b Y}{M}$

Question

Pure benzene freezes at 5.45°C at a certain place but a 0.374 m solution of tetraethylchloroethane in benzene freezes at 3.55°C . The K_f for benzene is-

- A** $5.08\text{ K kg mol}^{-1}$
- B** 508 K kg mol^{-1}
- C** $0.508\text{ K kg mol}^{-1}$
- D** $50.8^{\circ}\text{C kg mol}^{-1}$

Question

Addition of 0.643 g of a compound to 50 mL of benzene (density 0.879 g/mL) lowers the freezing point from 5.51°C to 5.03°C. If K_f for benzene is 5.12 K mol kg. Calculate the molecular weight of the compound.

- A** 153
- B** 167
- C** 156
- D** 187

Question

An aqueous solution of a non-electrolyte solute boils at 100.52°C . The freezing point of the solution will be-

- A** 0°C
- B** -1.86°C
- C** 1.86°C
- D** None of these

Question

At 100°C vapour pressure of aqueous solution of glucose is 750 mm of Hg. Then find the mole fraction of the solute.

Question

The vapour pressure of aqueous solution of sucrose is 732 mm of Hg. Then find the b.p. of the solution. If the vapour pressure of pure H_2O is 760 mm of Hg. (K_b of water = $0.52 \text{ kg K mol}^{-1}$)

Question

0.001 M glucose solution is present in a beaker. Then find its osmotic pressure at 27°C temperature.

Question

0.001 M glucose solution is present in a beaker. Then find its osmotic pressure at 27°C temperature.

Question

0.5 g solute is dissolved in 20 mL of solution then the osmotic pressure of solution is 7.6 mm of Hg at 27°C. Find molecular weight of solute.

Question

The osmotic pressure of 0.2 molar solution of urea at 300 K ($R = 0.082 \text{ litre atm mol}^{-1} \text{ K}^{-1}$) is-

- A** 4.92 atm
- B** 1 atm
- C** 0.25 atm
- D** 27 atm

Question

The osmotic pressure of 10%(w/v) aqueous solution of urea (π_1) is related to that of 10%(w/v) aqueous solution of glucose (π_2) as-

- A** $\pi_1 = \pi_2$
- B** $\pi_1 > \pi_2$
- C** $\pi_1 < \pi_2$
- D** $\pi_1 = 1 - \pi_2$

Question

A 5% (W/V) solution of cane sugar is isotonic with 0.877 % (W/V) of X. The molecular weight of substance X is-

A 58.98

B 119.96

C 95.58

D 126.98

Question

The Van't Hoff's factor (i) for a dilute aqueous solution of Na_2SO_4 is:

- A** $1 + \alpha$
- B** $1 - \alpha$
- C** $1 + 2\alpha$
- D** $1 - 2\alpha$

Question

The van't Hoffs factor of $\text{K}_4[\text{Fe}(\text{CN})_6]$ assuming 100% dissociation is:

- A** 5
- B** 4
- C** 11
- D** 6

Question

For the given electrolyte X_mY_n , the degree of dissociation ' α ' is given by ('i' is the Van't Hoff factor):

A $\alpha = \frac{i - 1}{m + n - 1}$

B $i = (1 - \alpha) + m\alpha + n\alpha$

C $\alpha = \frac{1 - i}{1 - m - n}$

D All of these

Question

Maximum lowering of vapour pressure is observed in the case of:

- A** 0.1 M glucose
- B** 0.1 M BaCl_2
- C** 0.1 M MgSO_4
- D** 0.1 M NaCl

Question

Which of the following aqueous solution will exhibit highest boiling point?

- A** 0.01M Na_2SO_4
- B** 0.015M $\text{C}_6\text{H}_{12}\text{O}_6$
- C** 0.01M Urea
- D** 0.01M KNO_3

Question

Which one of the following equimolal aqueous solutions will have maximum freezing point? (Assume equal ionisation in each case):

- A** $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$
- B** $[\text{CrCl}(\text{H}_2\text{O})_5]\text{Cl}_2\text{H}_2\text{O}$
- C** $[\text{CrCl}_2(\text{H}_2\text{O})_4]\text{Cl}_2\text{H}_2\text{O}$
- D** $[\text{Fe}(\text{H}_2\text{O})_3\text{Cl}_3]_3\text{H}_2\text{O}$

Question

The molal elevation constant of water is $0.52^{\circ}\text{C kg/mol}$. The boiling point of 1.0 molal aqueous KCl solution (assuming complete dissociation of KCl), therefore should be:

- A** 98.96°C
- B** 100.52°C
- C** 101.04°C
- D** 107.01°C

Question

What is the freezing point of a solution containing 8.1 g HBr in 100 g water assuming the acid to be 90% ionized (K_f for water = $1.86 \text{ K kg mole}^{-1}$):

- A** 0.85°C
- B** -3.53°C
- C** 0°C
- D** -0.35°C

Question

A 0.001 molal solution of $[\text{Pt}(\text{NH}_3)_4\text{Cl}_4]$ in water had a freezing point depression of 0.0054°C . If K_f for water is 1.80 , the correct formula for the above compound assuming its complete dissociation is:

- A** $[\text{Pt}(\text{NH}_3)_4\text{Cl}_3]\text{Cl}$
- B** $[\text{Pt}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}_2$
- C** $[\text{Pt}(\text{NH}_3)_4\text{Cl}]\text{Cl}_3$
- D** $[\text{Pt}(\text{NH}_3)_4\text{Cl}_4]$

Question

boiling point? (considering Molality $\approx$ M) Which one of the following solution has highest boiling point?

- A** 0.2 (M) Aqueous NaCl solution
- B** 0.5 (M) Aqueous Glucose solution
- C** 0.3 (M) Aqueous CaCl_2 solution
- D** 0.2 (M) Aqueous urea solution

Question

Select from the following that has highest osmotic pressure

- A** 0.1 N urea
- B** 0.1 N NaCl
- C** 0.1 N Na_2SO_4
- D** 0.1 N Na_3PO_4

Question

Which of the following solutions will have highest freezing point?

- A** 0.1 M KCl
- B** 0.1 M BaCl₂
- C** 0.1 M FeCl₃
- D** 0.1 M Fe₂(SO₄)₃

Question

What is ΔT_b for 0.2 molal aqueous solution of urea? (K_b of $H_2O = 0.52$)

- A** 0.104 K
- B** 101.04 K
- C** 0.506 K
- D** 5.06 K

Question

2 g of benzoic acid ($\text{C}_6\text{H}_5\text{COOH}$) dissolved in 25 g of benzene shows a depression in freezing point equal to 1.62 K. Molal depression constant for benzene is $4.9 \text{ K kg mol}^{-1}$. What is the percentage association of acid if it forms dimer in solution?

Question

0.6 mL of acetic acid (CH_3COOH), having density 1.06 g mL^{-1} , is dissolved in 1 litre of water. The depression in freezing point observed for this strength of acid was 0.0205°C . Calculate the van't Hoff factor and the dissociation constant of acid.

Question

Determine the amount of CaCl_2 ($i = 2.47$) dissolved in 2.5 litre of water such that its osmotic pressure is 0.75 atm at 27°C .

Question

Determine the osmotic pressure of a solution prepared by dissolving 25 mg of K_2SO_4 in 2 litre of water at 25°C , assuming that it is completely dissociated.

Question

Relative lowering in vapour pressure of a solution containing 1 mole K_2SO_4 in 54g H_2O is (K_2SO_4 is 100% ionised)

A $\frac{1}{55}$

B $\frac{3}{55}$

C $\frac{3}{4}$

D $\frac{1}{2}$

Question

Which of the following solutions will have highest boiling point:

- A** 0.1 M FeCl_3
- B** 0.1 M BaCl_2
- C** 0.1 M NaCl
- D** 0.1 M urea (NH_2CONH_2)

Question

0.1 Molal aqueous solution of an electrolyte AB_3 is 90% ionized. The boiling point of solution at 1 atm is [$K_b(H_2O) = 0.52K.kg.mol^{-1}$]

A 273.19K

B 274.92K

C 374.92K

D 373.19K

Question

24 g of $\text{C}_2\text{O}_2\text{H}_4$ is dissolved in 100 g water, in order to produce ice, mixture is cooled to -10°C , if $K_f = 1.86$, the amount of ice (in g) separated

- A** 25.6
- B** 50.5
- C** 75.6
- D** 20.8

Question

Amount of ice that will separate out on cooling a solution containing 50 Ethelene glycol in 200 gm water to -9.3°C will be
(K_f water = $1.86 \text{ K kg mol}^{-1}$)

- A** 38.7 g
- B** 3.87 g
- C** 83.7 g
- D** 8.37 g

Question

When mercuric iodide is added to the aqueous solution of KI, then the:

- A** Freezing point is raised
- B** Freezing point is lowered
- C** Freezing point does not change
- D** Boiling point does not change

Question

The degree of dissociation (α) of a weak electrolyte A_xB_y is related to van't Hoff factor (i) by the expression:

A $\alpha = \frac{i - 1}{(x + y - 1)}$

B $\alpha = \frac{i - 1}{(x + y + 1)}$

C $\alpha = \frac{(x + y - 1)}{i - 1}$

D $\alpha = \frac{(x + y + 1)}{i - 1}$

Question

List-I

- (A) Van't Hoff Factor, i
- (B) k_f
- (C) Solutions with same
- (D) Azeotropes

List-II

- (I) Cryoscopic constant
- (II) Isotonic solutions
- (III) Normal molar mass
Abnormal molar mass
- (IV) Solutions with same composition of vapour above it

Choose the correct answer from the options given below:

- A** (A)-(III), (B)-(I), (C)-(II), (D)-(IV)
- B** (A)-(III), (B)-(II), (C)-(I), (D)-(IV)
- C** (A)-(III), (B)-(I), (C)-(IV), (D)-(II)
- D** (A)-(I), (B)-(III), (C)-(II), (D)-(IV)

Question

Solution of 12 g of non-electrolyte (A) prepared by dissolving it in 1000 mL of water exerts the same osmotic pressure as that of 0.05 M glucose solution at the same temperature. The empirical formula of A is CH_2O . The molecular mass of A is _____ g. (Nearest Integer)

Question

If the boiling points of two solvents X and Y (having same molecular weights) are in the ratio 2 : 1 and their enthalpy of vaporizations are in the ratio 1 : 2, then the boiling point elevation constant of X is m times the boiling point elevation constant of Y. The value of m is ____ (Nearest Integer)

Question

Henry's constant (in kbar) for four gases α , β , γ and δ in water at 298 K is given below:

	α	β	γ	δ
K_H	50	2	2×10^{-5}	0.5

(density of water = 10^3 kg m^{-3} at 298 K)

This table implies that:

- A** α has the highest solubility in water at a given pressure
- B** solubility of γ at 308 K is lower than at 298 K.
- C** The pressure of a 55.5 molal solution of δ is 250 bar.
- D** The pressure of a 55.5 molal solution of γ is 1 bar.

Question

The number of pairs of the solution having the same value of the osmotic pressure from the following is _____.
(Assume 100% ionization)

- A** 0.500 M $\text{C}_2\text{H}_5\text{OH}$ and 0.25 M KBr (aq)
- B** 0.100 M $\text{K}_4[\text{Fe}(\text{CN})_6]$ (aq) and 0.100 M $\text{FeSO}_4(\text{NH}_4)_2\text{SO}_4$ (aq)
- C** 0.05 M $\text{K}_4[\text{Fe}(\text{CN})_6]$ (aq) and 0.25 M NaCl (aq)
- D** 0.15 M NaCl (aq) and 0.1 M BaCl_2 (aq)
- E** 0.02 M KCl $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$ (aq) and 0.05 M KCl (aq)

Question

The value of Henry's law constant for some gases at 293 K is given below.
Arrange the gases in the increasing order of their solubility.

He : 144.97 kbar, H₂ : 69.16 kbar,

N₂ : 76.48 kbar, O₂ : 34.86 kbar

A He < N₂ < H₂ < O₂

B O₂ < H₂ < N₂ < He

C H₂ < N₂ < O₂ < He

D He < O₂ < N₂ < H₂

Question

0.004 M K_2SO_4 solution is isotonic with 0.01 M glucose solution. Percentage dissociation of K_2SO_4 is ____
(Nearest Integer)

Question

Two solutions A and B are prepared by dissolving 1 g of non-volatile solutes X and Y, respectively in 1 kg of water. The ratio of depression in freezing points for A and B is found to be 1 : 4. The ratio of molar masses of X and Y is

A 1 : 4

B 1 : 0.25

C 1 : 0.20

D 1 : 5

Question

Solute A associates in water. When 0.7 g of solute A is dissolved in 42.0 g of water it depresses the freezing point by 0.2°C . The percentage association of solute A in water, is

[Given: Molar mass of A = 93 g mol^{-1} . Molal depression constant of water is 1.86 kg mol^{-1}]

A 50%

B 60%

C 70%

D 80%

Question

The elevation in boiling point for 1 molal solution of non volatile solute A is 3K. The depression in freezing point for 2 molal solution of A in the same solvent is 6 K. The ratio of K_b and K_f i.e K_b/K_f is 1 : X. The value of X is _____. [Nearest Integer]

Question

The depression in freezing point observed for a formic acid solution of concentration 0.5 mL L^{-1} is 0.0405°C . Density of formic acid is 1.05 g mL^{-1} . The Van't Hoff factor of the formic acid solution is nearly: (Given for water $K_f = 1.86 \text{ K kg mol}^{-1}$)

- A** 0.8
- B** 1.1
- C** 1.9
- D** 2.4